

Implementation of the Dynamic Time Warping Algorithm for Bioacustic Detection

by

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> *A very long quote*
> *from someone very important.*
> – Mr M

Thank you to Leanne for helping to prepare this template.

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Abstract

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Nomenclature

Acronyms and Abbreviations

DCT	Discrete Cosine Transform
DTW	Dynamic Time Warping
FT	Fourier Transform
LPC	Linear Predictive Coding
LPCC	Linear Predictive Coding Coefficients
MFCC	Mel-Frequency Cepstral Coefficients
SNR	
STFT	Short Time Fourier Transform

Definitions

F

F1-score i

frame duration i

H

hamming window i

P

precision i

R

recall i

S

sampling frequency i

spectral leakage i

Chapter 1

Introduction

1.1 Background

Since the late 1990s there have been indications that whales have been struck by shipping vessels.[1] Today this has become a larger concern, as vessel speeds and traffic has substantially increased. Bryde's whales are found to migrate in smaller regions instead of the usual large-scale migrations of other whale species. Such a whale colony in Fasle Bay, South Africa has also been placed in possible danger due to fishing activities in their migration region.[2]

Detection and classification systems for whale calls can be used to monitor the migration patterns of specific whale colonies throughout the year. The information gained from such analysis can help determine when vessel traffic should be diverted for the safety of a whale population. These types of systems have been implemented on different whale colonies, but the migration patterns of the False Bay colony has not yet been thoroughly documented.[3] Due to the variations in call types between different colonies of some whale species (killer whales[4], blue whales[5]), models trained for other Bryde's whale colonies would not be completely trustworthy for the detection of calls within the False Bay Bryde's Whale colony as universiality of calls cannot be assumed. Therefore, a model trained specifically for the False Bay colony would be beneficial.

Dynamic time warping (DTW) has been shown to be an effective whale classification method on different whale colonies.[6] Given the limited labeled data for the False Bay colony, DTW is considered as it requires less input data than other detection algorithms.[7]

1.2 Problem Statement

This project will determine whether using DTW using vector features is able to effectively detect Bryde's Whale calls within the False Bay region. Three feature extraction methods

will be compared. These will be Short time fourier transform (STFT), Mel-frequency cepstral coefficients (MFCC) and Linear predictive coding (LPC).

1.3 Objectives

1. Implement a DTW detection system using three different feature extraction methods (STFT, MFCC and LPC). This shows whether the effectiveness of the DTW system is independent from the feature extraction method.
2. Calculate the precision, recall and F1-score for each model.
3. Compare results and determine whether DTW is overall a good detection method and which feature extraction method is best suited.

1.4 Scope

Elements not considered within this project include the classification of call types after detection and the detection of labeled tonal downsweep calls.

Call type classification was not considered due to discontinuities in how calls were labelled. These discontinuities arose due to a variety of human labelers for different sections of data and the presence of similar frequency components across different call types.

Sections of audio containing labeled tonal downsweep calls were ignored due to a lack of data across multiple sections of recordings for this call type.

1.5 Layout of report

* Chapter 2 provides any background information needed to understand how to solve the given problem as well as any helpful literature of previous solutions to the problem.
* Chapter 3 provides background information on the dataset used for this project as well as explaining the splitting of data for the project.
* Chapter 4 gives a detailed design of the entire project.
* Chapter 5 provides evidence of results obtained after the system had been designed as well as descriptions of what the results show.
* Chapter 6 contains all concluding thoughts on the results as well as improvements and recommendations for the project.

Chapter 2

Literature

2.1 Bryde's Whales

Bryde's Whales predominently live in tropical or subtropical environments across the world. The different call types, rates, frequencies and harmonics of these whales vary depending on geographical location of the colony. Generally Bryde's Whales have calls containing frequencies from as low as 42 Hz to as high as 1kHz.

In the dataset used for this project there were four main call types present. These are referred to as single-tones, multi-tones, burst-tonals and tonal downsweeps.

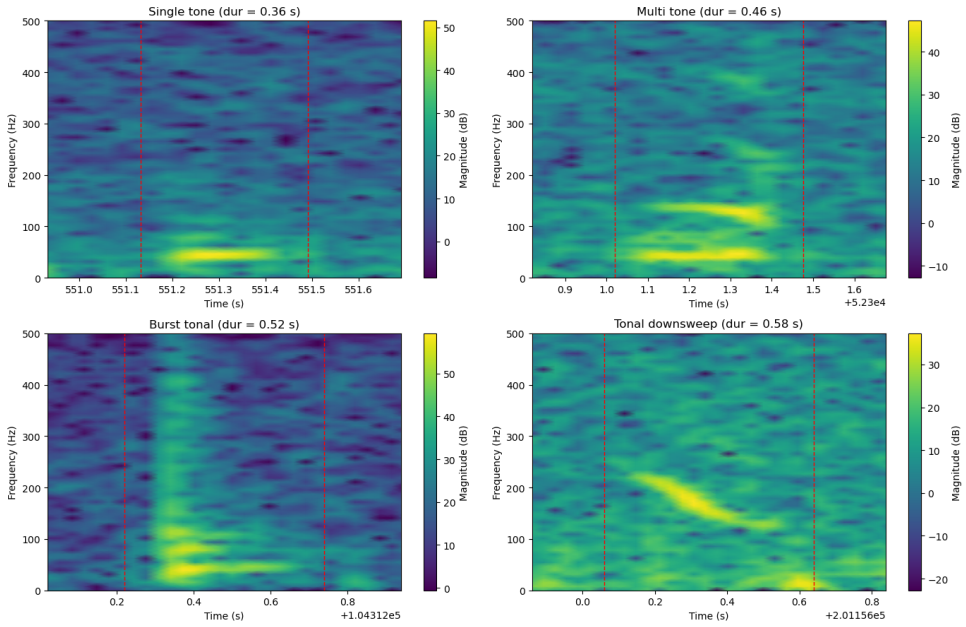


Figure 2.1: Examples of how each call type looks: single-tone, multi-tone, burst-tonal and tonal downsweep.

The single-tone calls were charecterized by a singular frequency at around 42 Hz. Multi-tones also contained the 42 Hz frequency, but there were one or two other frequencies above this with a maximum frequency component at about 200 Hz. Burst-tonals also has the multiple frequency characteristic of the multi-tones, however the lower frequencies, 42 - 200 Hz, were held for longer than the higher frequencies, 200 - 500 Hz. The tonal downsweep was the only call type not characterized by the 42 Hz frequency. It was a sweeping of frequencies through time starting at around 200 Hz and ending at around 100 Hz.

In accordance with the original detected calls the average call length of each call type was around 0.3 seconds with a maximum duration of about 2.5 seconds. In other studies of Bryde's Whales calls in different geographical areas a maximum duration of arount 7.0 seconds was identified.[2][8]

Decisions made in the project pertaining to frequencies and durations falling within the detection limits were made based on the above information.

2.2 Dynamic Time Warping

Dynamic time warping (DTW) is an algorithm for detecting the similarity of two signals, $X = [x_1, x_2, \ldots, x_n]$ and $Y = [y_1, y_2, \ldots, y_m]$. This measure of similarity is achieved by calculating an optimal alignment cost between the two signals of length $N \in \mathbb{N}$ and $M \in \mathbb{N}$ respectively. In terms of feature vectors such as the short time fourier transform (STFT), mel-scake frequency cepstral coefficients (MFCC) or linear predictive coding coefficients (LPCC) this means that each frame making up one signal is compared to each frame making up another signal. Using an alignment cost the best alignment for each frame in one to the other signal is chosen and an overall DTW cost is calculated to show how similar the overall signals are. Figure 2.2 shows that such an optimal alignment path may mean that a single frame in signal X may produce the lowest cost against more than one frame in signal Y and vice versa.[9][10]

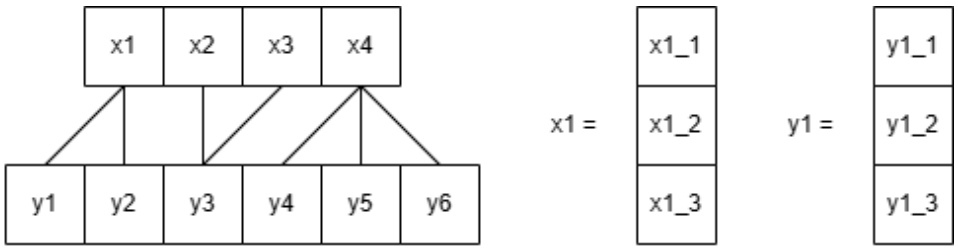


Figure 2.2: An example of the optimal warping path between two vector features, X with length N = 4 and Y with length M = 6

The overall cost needed to determine how similar the two signals are is calculated via a

cost matrix. Cost matrix, $C_{N+1,M+1}$ for signals X and Y, is initialised such that $C_{i,0} = \infty$ for i = 1 to M, $C_{0,j} = \infty$ for j = 1 to M and $C_{0,0} = 0$.

The formula used to fill each cell of the cost matrix is:

$$C_{i,j} = \text{dist}(x_i, y_j) + \text{Q} \qquad \text{with} \quad \text{Q} = \min \begin{cases} C_{i-1,j-1} & \text{(match)} \\ C_{i-1,j} & \text{(insertion)} \\ C_{i,j-1} & \text{(deletion)} \end{cases} \tag{2.1}$$

where $dist(x_i, y_j)$ represents the difference between X_i and Y_j as a distance. For vector signals this distance is calculated between each frame i of signal X and each frame j of signal Y. This can be done using various different techniques including cosine, euclidean or dot product similarity. For this project only the cosine similarity and euclidean distance metrics were used.[10]

Euclidean distance is sensitive to the magnitude and direction of the vectors for which it is calculated. In basic terms it is the absolute difference between two vectors.[11]

$$d(\mathbf{x_i}, \mathbf{y_j}) = \sqrt{(\mathbf{x_i}_1 - \mathbf{y_j}_1)^2 + (\mathbf{x_i}_2 - \mathbf{y_j}_2)^2 + \cdots + (\mathbf{x_i}_n - \mathbf{y_j}_n)^2} \tag{2.2}$$

Cosine similarity is only sensitive to the direction of the vectors for which it is calculated. In basic terms it is the angle between the two vectors.[11]

$$\cos(\mathbf{x_i}, \mathbf{y_j}) = \frac{\mathbf{x_i} \cdot \mathbf{y_j}}{\|\mathbf{x_i}\| \, \|\mathbf{y_j}\|} \tag{2.3}$$

Q is the minimum value of three possible options namely match, insertion or deletion. Match is the value in the matrix cell diagonally down and left from the current cell for which the cost is being calculated. Insertion is the value in the matrix cell directly below the current cell and deletion is the matrix cell directly to the left of the current cell.

The cost which is then calculated for the current cell, $C_{i,j}$, is a summation of the distance between the feature frames i and j in signals X and Y respectively as well as the lowest accumlated cost up until that point. This means that the final cell reached within the cost matrix, $C_{N+1,M+1}$, will contain the lowest overall cost for the similarity of signals X and Y. The lower the final cost, the higher the similarity between the two signals. The example shown in Figure 2.3 is shown if X and Y are chosen to be scalar signals, however the same logic is used for when X and Y are vectors.[10]

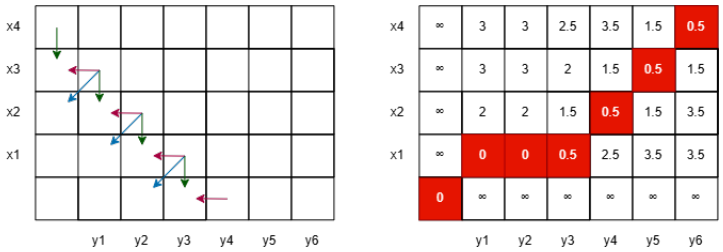


Figure 2.3: An example for the three possible options for min as well as an example of the optimal warp path found for signals X and Y assuming X=[0, 2, 1, 0] and Y=[0, 0, 0.5, 2, 1, 0]

The general case of DTW finds global alignment between the two signals. This alignment is found using the warping path.The warping path is the path of the best alignment between signals X and Y within the cost matrix, as shown in Figure 2.3. This path, W, is a sequence of matrix co-ordinates, (i, j), with a total length k.[12]

The boundary condition for global alignment ensures the final path starts and ends with the start and end frames of each respective signal.

$$i_1 = 1, i_k = N \quad \text{and} \quad j_1 = 1, j_k = M \tag{2.4}$$

The monotonicity condition ensures that each new point in the warping path is a step forward in time from the previous step.

$$i_{t-1} \leq i_t \quad \text{and} \quad j_{t-1} \leq j_t \tag{2.5}$$

The continuity condition ensures the step size of the warping path is only one unit in time.

$$i_t - i_{t-1} \leq 1 \quad \text{and} \quad j_t - j_{t-1} \leq 1 \tag{2.6}$$

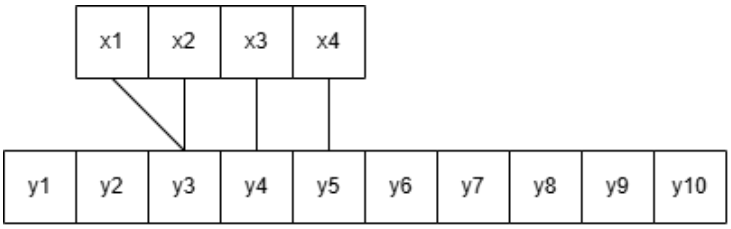


Figure 2.4: Example of a subsequence DTW alignment vetween two signal $X = [x_1, x_2, x_3, x_4]$ and $Y = [y_1.y_2, y_3, y_4, y_5, y_6, y_7, y_8, y_9, y_10]$

Subsequence DTW is a version of the DTW algorithm that alters the boundary condition such that global alignment is no longer necessary. Instead the entirety of signal X is now matched with a subsequence of signal Y. The new boundary condition allows the

warping path to start and end at any point within the first and final row of the matrix respectively.

$$i_1 = 1, i_k = N \quad \text{and} \quad 1 \leq j_1 \leq j_k \leq M \tag{2.7}$$

In global alignment the final cell in the cost matrix, (N+1, M+1), is considered the overall optimal cost for those two signals. With subsequence DTW the entire final row of the matrix is checked to find the lowest cost. The position of that cost shows where the end point of the alignment detected via subsequence DTW is.[13]

2.3 Feature Extraction Methods

2.3.1 Short Time Fourier Transform

The short time fourier transform (STFT) is used to analyse the change in frequency of a signal over time. The STFT of a segment of audio is calculated by multiplying the signal in the time domain, $x(t)$ with a window $w(t-\tau)$ which is sweeped across said signal. The fourier transform (FT) of each window is calculated and used to determine the frequency information for that window. Thus, the STFT provides the change in frequency for a segment of audio throughout time.[14][15]

The mathematical formulation of STFT is:

$$X(\tau, \omega) = \int_{-\infty}^{\infty} x(t)\, w(t-\tau)\, e^{-j\omega t}\, dt, \tag{2.8}$$

where τ is the position of the window as it shifts through time at step-size H and ω is the frequency being analysed.

The frequency resolution of the STFT is given as:

$$resolution = \frac{f_s}{L} \tag{2.9}$$

where f_s is the sampling frequency of the signal and L is the length of the window over which the FT is taken.

This means that the STFT has a trade-off between time and frequency resolution.

If L is kept short, time resolution is high since smaller segments of time are being used for one FT analysis. However, frequency resolution is then decreased since too little time passes for the FT to distinguish between frequencies spaced closer together.

If L is kept long, time resolution drops. This means that fast changes in frequency will be missed. However, frequency resolution increases and it becomes easier to distinguish between differrent frequency bands.

Window overlap is another consideration when using STFT. Increased overlap creates

a smoother representation of the changes in frequency occuring over time since information at the edges of the window is no longer lost. The trade-off with overlap is the increase in computational complexity that will occur due to smaller and thus more windows of a signal being analysed.[9][15]

2.3.2 Mel-Frequency Cepstral Coefficients

Mel-frequency cepstral coefficients (MFCC) is a common choice for feature extraction when working on aucoustic data. This is because it is set up to percieve frequencies as they are percieved by the human ear. That means the lower frequencies are given a higher resolution than the high frequencies.[16] Features are initially extracted using the fast fourier transform (FFT). After this they are passed through a mel-scale filter bank. The formula for the mel-frequency calculation is:

$$Mel(f) = 1127 \ln\left(1 + \frac{f}{700}\right) \tag{2.10}$$

with $Mel(f)$ as the frequency value on the mel scale and f as the actual frequency in Hz.[14]

This bank is made up 20 to 40 triangular filters which are placed, smallest width to largest width according to the mel scale, along the frequency spectrum. The standard amount of triangular filters chosen is 26. This results in 26 coefficients for the entire range of frequencies being analysed.[17][16]

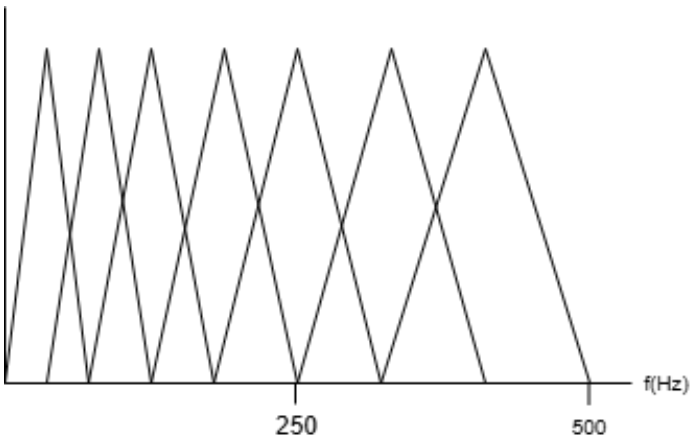


Figure 2.5: The mel scale filter bank is made up of 20 to 40 triangular filters arranged according to the mel scale.

The log of these coefficients is then taken and finally the discrete cosine transform (DCT) is used to decorrelate the filter bank coefficients. That means any patterns linking different coefficients to each other are removed such that the coefficients become independent. The

final number of cepstral coefficients returned by the DCT is typically 12 or 13 since the most important information is kept there. The formula use to perform the DCT is:

$$MFCC_i = \sum_{k=1}^{K} X_k \cos\left(\frac{i(k-0.5)\pi}{K}\right), \quad i = 1, 2, \ldots, K \tag{2.11}$$

with i as the index of the current coefficient, K as the total number of triangular filters used, k as the specific filter bank at that point in time and X_k as the energy after log was applied to the mel coefficients.[14]

The first and second derivatives, often referred to as deltas, of the cepstral coefficients are often used in feature extraction. They show the change in frequency between cepstral coefficients. This can be vital information when analysing how frequency changes across time since the MFCC process tends to lose this information. The formula for the derivatives is:

$$x_t^{(d)} = \frac{\sum_{m=-M}^{M} m \cdot x_{t+m}^{(d-1)}}{\sum_{m=-M}^{M} m^2} \tag{2.12}$$

with t as the specific point in time, d as the order of derivative and m as the frame offset.[18][19][20]

2.3.3 Linear Predictive Coding

Linear predicitve coding (LPC) is a straightforward mathematical method commonly used for feature extraction. LPC focuses on the method of approximating a sample of a speech signal using a combination of n samples from the past. This means that for each sample for which there are p available samples before it an approximation is made in the form of the equation:

$$\tilde{s}(n) = \alpha_1 s(n-1) + \alpha_2 s(n-2) + \cdots + \alpha_p s(n-p) \tag{2.13}$$

where $\tilde{s}(n)$ is the approximated speech sample and $\alpha_1, \alpha_2, \ldots, \alpha_p$ are the coefficients for that approximation. The p coefficients used are constant over all samples within the given frame.

For each sample with an approximation an error is calculated between the real sample and the approximated sample.

$$e_n = s(n) - \tilde{s}(n) \tag{2.14}$$

The sum of the squares of all these errors is minimised to find the p constant coefficients. The equation for this is:

$$E_n = \sum_m e_n^2(m) \tag{2.15}$$

where m is the sample for which the error $e_n(m)$ was calculated.

The feature extracted via this method is made up of the p LPC coefficients which were calculated to minimise the mean square error. These coefficients tend to be unstable, therefore cepstral coefficients can be calculated from these raw LPC coefficients giving linear predictive coding coefficients (LPCC). LPCC gives the coefficients of the fourier transform (FT) representation of the magnitude spectrum and are more reliable than the LPC representation.

LPCC can be calculated from the LPC results through recursion. The formula for that is:

$$\begin{aligned} c_0 &= \ln \sigma^2 \\ c_m &= a_m + \sum_{k=1}^{m-1} \left(\frac{k}{m}\right) c_k a_{m-k}, \qquad 1 \leq m \leq p \\ c_m &= \sum_{k=1}^{m-1} \left(\frac{k}{m}\right) c_k a_{m-k}, \qquad m > p, \end{aligned} \tag{2.16}$$

where σ^2 is the LPC gain term.

The final feature extracted using LPCC contains p coefficients for a single frame with $p = 0$ containing the magnitude information.[21][14]

2.4 Similar Studies

2.4.1 Classification of Killer Whale Vocalizations using DTW

An article, "Automatic classification of killer whale vocalizations using dynamic time warping", written by Judith, C. and Patrrick J.O. Miller evaluated four different variations of the dynamic time warping (DTW) algorithm to classify biphonic killer whale vocalizations using fundamental frequency contours.

Due to the biphonic nature of these calls the low-frequency component and high-frequency component for each call was extracted for seperate analysis. Four different variations of the DTW algorithm with various different constraints were used to evluate against each other.

The final result of the paper was that the method referred to as "Sakoe-Chiba" on the LPC performed the best with a 90% agreement with the labeles selected by human classifiers.

This study did show that the use of DTW in bioacustic analysis is a good option.

However, it did not consider the ability of DTW to detect call in a continuos segment of bioacustic data as all calls which were classified had already been selected and segmented to the pitch contours of the vocalization. This study also did not consider the impact of low frequency analysis using DTW as killer whale vocalizations do not tend to fall below 500 Hz.[22]

2.4.2 Detection of Template-Based Detection Algorithms for Bryde's Whales

The article, "Analysis of Template-Base Detection Algorithms for Inshore Bryde's Whale Short Pulse Calls" written by Olayinka O. Ogundile and Daniel J. J. Versfeld analysed two template based detection algorithms to detect short pulse calls for Bryde's Whales in False Bay, South Africa. The two methods tested were dynamic time warping (DTW) and linear predictive coding (LPC). In their analysis they used two seperate sample sets with different signal-to-noise ratio (SNR) values to also evaluate the performance of these techniques in the presence of noise. The input features for these models were the time-domain representation of the signals.

The final results of the article found that both LPC and DTW performed well in high and low SNR segments. LPC performed the overall best in low noise segments.

This project provides insight into the use of template based detection systems for Bryde's Whales. This is relevant since DTW which is handled in this article is the chosen detection mechanism for this project. While LPC is also used in the article, its use is vastly different to the use within this project. LPC is used as a detection algorithm and tested on its ability to act as such. In this project LPC is not used as a detection algorithm, but instead linear predictive coding coefficients (LPCC) is used as a feature extraction method. The detection method focuses on the error component of the LPC equation while the feature extraction method focuses on the coefficients calculated for the approximation of the real signal. Another difference between this project and the article is that this project exclusively uses the time domain representation of the signal as input.[23]

2.4.3 Detection of real-time fin and blue whale calls

The article, "Automatic recognition of fin and blue whale calls for real-time monitoring in the St. Lawrence" by Xavier Mouy, Mohammed Bahoura and Yvan Simard tested detection models on real-time data for fin and blue whales. The detection methods used were dynmaic time warping (DTW) and vector quantization (VQ). This article focused on low frequency whale calls.

Instananeous frequency, speed and acceleration was extracted as features. It was found that due to low frequency components from background elements such as ships the detection of false positives (FP) increased.

In terms of this project the article shows how low frequency detections can be affected by low frequency background noise. This article also considered real-time audio detection which is much closer in similarity to detection within a large audio file such as what was done in this project. This article did not use the same feature vectors as the project and did not consider their effect on DTW performance.[24]

2.4.4 Hidden Markov Model for Bryde's Whale short pulse calls

The article "A hidden Markov model with selective time domain feature extraction to detect inshore Bryde's whale short pulse calls" by O.O. Ogundile, A.M. Usman, O.P. Babalola and D.J.J. Versfeld compares a time domain feature extraction technique (MAP) used for hidden markov models (HMM) against mel-frequency cepstral coefficients (MFCC) and linear predictive coding (LPC), frequency domain techniques. The model is used for detection of Bryde's Whale calls in False Bay, South Africa.

These three HMM models are used on 2 hours of continuos recordings to detect these calls. This is similar to the continuos recording detection done in this project.

The MAP-HMM model performed the best with MFCC performing second best and LPC performing the worst. It was found that the MFCC and LPC models were sensitive to background noise.

While this does test detection on a continuos segment of audio, this article focuses on HMM detection and not DTW. It also does test MFCC feature extraction method as well as the LPC method. Considering the fact that the LPC features were not converted to linear predictive coding coefficients (LPCC) this does not fully align with this project.[25]

2.4.5 Speech Recognition using DTW

Th goal of the article "Speech recognition using Dynamic Time Warping (DTW)" by Yurika Permanasari, Ermin H. Harahap and Erwin Prayoga Ali was to analyse the effectiveness of dynamic time warping (DTW) in matching speech patterns. Variabilities were included into the study via pronounciation timing, volume and speaker identity.

Features were extracted from the recorded audio. Some audio was used as templates and others as test data. DTW was then used to test which of the variations impacted DTW distances the most. The main tests were keeping the pronounciation time the same with the same volume, different pronounciation time with the same volume and same pronounciation time with different volume.

It was determined that DTW was an effective pattern recognition algorithm and that pronounciation timing had a larger impact on DTW alignment than volume.

This study also uses DTW to recognise patterns in audio, however this focuses on speech recognition and the effects of the abilities of variances in that recognition. It

does not focus on different feature vectors or detection of low frequency whale calls in a continuos recording.[26]

Chapter 3

Dataset

3.1 Background

The dataset used contains three 72 hour recordings, recorded in False Bay, near Gordons Bay, between 22 April 2023 and 22 June 2023. The recordings were taken at a depth of 20 to 30 meters at the coordinates, S 34.17°, E 18.76°.

The device used for the recordings was a single omnidirectional hydrophone, designed and built by a Stellenbosch University research group from the Electrical and Electronic Engineering Department.

The recorded data has a sampling rate of 8kHz and a 16-bit resolution. The data was extracted as a 72 hour wav file from an SD card used by the recording device.[27]

3.2 Labeling

Detection tables, containing start and end times of calls within each of the three file, were provided from a previous research project which used a band-limited energy detector to flag calls. This method unfortunately did not detect all the calls within the audio files and none of the calls were labeled as a specific type of call.

Another detection table was procured which contained more detections for the first 72 hour file and contained call labels for all three files. This meant that while all three files contained labeled calls only one contained enough labeled calls to use the entire file without expecting an excessive amount of unlabeled calls.[27]

3.3 Data Splitting

For this project the data was split in three sections, template data, hyperparameter tuning data and evaluation data. Each of the 72 hour audio files was chosen for a different

section. Given that only the first of the three 72 hour audio files was well labeled, it was decided that this would be used as the evaluation data. The second audio file was used for hyperparameter tuning and the third was used to select templates. Labeled calls for the down sweep call type were only found in the audio file used for evaluation. Therefore, due to lack of data, this call type was ignored.

Table 3.1: Dataset counts by date and category

Recording Start Date	Call Count				Total Calls
	Single-Tone	Multi-Tone	Burst-Tonal	Tonal Downsweep	
22 April 2023	1015	452	171	164	1802
26 April 2023	242	91	9	0	342
29 April 2023	32	25	11	0	164

3.3.1 Template Selection

The audio file which was recorded starting on 29 April 2023 was used to select templates. These chosen templates were used as the baseline for DTW comparisons.

For the three considered call types, single-tones, multi-tones and burst-tonals, the maximum number of templates per call type which could be extracted such that each call type had the same amount of templates was 10. The limiting call type was burst-tonals. In Table 3.1 it can be seen that there were 11 available burts-tonals. However, it was decided to make 10 the maximum number of templates per call type.

To ensure that the 10 templates chosen were representative of varying signal-to-noise ratios SNR templates were chosen by looking at spectrograms of labeled calls throughout the audio file. The visual inspection was also necessary since some of the calls labeled as single-tones looked more like multi-tones and some of the multi-tones could actually be classified as burst-tonals. This overlap in possible classification was due to the variety of human labelers and the common frequency components between call types.

3.3.2 Hyperparameter Tuning Template Selection

The audio file which was recorded starting on 26 April 2023 was used to select templates used for hyperparameter tuning, referred to as calibration templates. These calibration templates were used to determine below which DTW cost threshold a segment of audio could be considered a call. For the calibration templates 10 templates were chosen for each call type. This was again limited by the burst-tonals maximum number of templates. In this file there were only 9 labeled burst-tonals, but when selecting multi-tones it was

found that a call originally labeled as a multi-tone could in fact be seen as a burst-tonal. The number of each call type labeled per file can be seen in Table 3.1.

Besides these 30 calibration templates, 10 for each call type, 100 background calibration templates were also selected. Within each audio file most of the time was made up of background noise and only some parts actually contained calls. This was simulated by selecting many more background calibration templates that call-type calibration templates. Since there is no table labeling background noise background calibration templates of varying sizes were extracted from the file and 100 templates were manually selected. The reason for the manual selection process links with the unreliability of the labeled data. The audio file used for selection of calibration templates was only labeled by the band-limited energy detector and thus did not label all the calls within the file. This means that some of the samples extracted as background audio actually contained calls that had never been labeled. Manually selecting each template ensured that all the templates chosen only contained background noise.

3.3.3 Evaluation Data Selection

The audio file recorded starting on 22 April 2023 was used for the final evaluation of the whale call detection model. No templates were extracted, instead the entire file was sweeped to search for calls.

As seen in Table 3.1 this file was the only file which contained tonal-downsweep calls. Due to the limited amount of these types of calls all the labeled tonal-downsweeps within this file was ignored while testing.

Chapter 4

System Design

4.1 Overview

The project was coded in Python. The functionality of each section of the project was placed in relevant python scripts. This was imported to the relevant jupyter notebook which was used for that section.

4.2 Feature Extraction Methods

The raw audio recorded by the hydrophone was stored in wave files with a sampling frequency of 8kHz. Before extracting feature vectors fromthe recordings the audio was first downsampled to have a 1 kHz sampling rate. This helped remove unecessary frequency information as the Nyquist Frequncy ($\frac{f_s}{2}$) for a 1kHz sampling rate is 500 Hz.[28]

Most of the Bryde's Whale vocalizations occur below 500 Hz. Thus, this sampling rate also ensures no relevant information was lost before analysis.[2]

4.2.1 Short Time Fourier Transform

The signal.stft function from the SciPy library was used to calculate the short time fourier transform (STFT) feature vectors. Window length L, as referenced in Eq (2.9), is calculated by multiplying f_s, the sampling of the given audio segment, with the chosen STFT frame duration. No set frame duration was selected as the optimal frame duration was chosen during the hyperparameter selection stage of this project.

The window ovelrap H, as referenced in Eq (2.9), is calculed by multiplying L with the chosen overlap percentage. For this project the overlap percentage was chosen as 0,75. The high overlap percentage produced a smooth spectrogram when viewing the time–frequency representation of the given audio segment.A hamming window was chosen to decrease any

spectral leakage caused by the finite length of the windows used to extract the feature vectors.[15][29]

4.2.2 Mel-Frequency Cepstral Coefficients

The feature.melspectrogram and feature.mfcc functions from the Librosa library were used to calculate the cepstral coefficients for the mel-frequency cepstral coefficients (MFCC) feature extraction implementation. The window length for the fourier transform (FT) required is calculated by multiplying f_s with the chosen frame duration which is optimally selected during hyperparameter selection. Unlike the STFT function librosa.feature.mfcc requires the hop length and not the overlap length. This is calculated by subtracting the window overlap, H, from the frame length, L. H is calculated by choosing the overlap percentage to be 0,75 and multiplying this with L. A hamming window is used to decrease spectral leakage.

The Librosa function used for calculating MFCC automatically uses discrete cosine transform (DCT) type 2 and ortho normalization. These settings mean that the bottom row of the output matrix, c_0, represents the sum of the log-spectrum energy. It is common practice in speech analysis to remove coefficient 0, as it contains the total energy of the signal and can dominate distance calculations in subsequent analysis. For this project, only coefficients 1 to 14 were kept. This ensured that the analysis only considered the overall shape of the log spectrum and not the total energy of the signal. The number of melspectrogram filters was chosen as 26, which is common practice in audio processing. 14 final coefficients were calculated. 13 are typically used, therefore with the lowest coefficient 0 removed 14 were calculated.[30][31][32]

The feature.delta function from the librosa library was used to calculate the first and second derivitives for MFCC. The feature returned when using the first derivitive was a concatenation of the original MFCC feature with the first derivitive of that feature vector. This gave a feature with 26 coeficients, the original 13 coefficients with 13 added from the first derivitive. Similarly for the second derivitive the MFCC and first derivitive concatenation was concatenated with the new second derivitive. This gave a 39 coefficient feature vector.[33][34]

4.2.3 Linear Predictive Coding

The util.frame and lpc functions from the Librosa library were used to calculate the linear predictive coding (LPC) features. The STFT and MFCC functions created the frames within their functions and thus it was not necessary to frame the audio before using it as an input. The LPC fuinction takes in a sequence of frames to calculate the approximation coefficients. The function util.frame was used with a frame duration selected during hyperparameter selection and a hop length of 0.25 of the frame duration, $1 - 0.75$. A

hamming window was also applied to the frames before using the lpc function.[35][36]

The higher the LPC order the more detailed the approximation becomes. Considering that background noise could have a negative effect on the coefficient outcomes if the approximation becomes too detailed, an order of 4 was chosen.[21]

Due to the instability of the LPC extraction method the LPC feature was converted to linear predictive coding coefficients (LPCC). This was done by manually typing out the recursive formula Eq 2.16. The number of LPCC coefficients typically chosen is approximately $\frac{3}{2}p$. For p = 4 it was chosen that around 6 LPCC coefficients should be calculated. After testing different numbers of LPCC coefficients it was found that 8 LPCC coefficients gave the best F1-scores during hyperparameter selection. Therefore, 8 was the chosen number of LPCC coefficients used.[21]

4.3 DTW methodology

The sequence.dtw function from the Librosa library was used to calculated the DTW cost matix between a template and an audio segment. The distance matrix needed as input for the function was calculated using the spacial.distance.cdist function from the SciPy library.

The distance function was given the template feature and the audio segment feature extracted either by STFT, MFCC or LPC as well as the metric specified for each extractor.The two metrics considered were cosine similarity and euclidean distance. As shown in Eq.(2.3) and Eq.(2.2), cosine similarity calculates the distance in angle between two points and euclidean distance calculates the acual distance between the two point. For the STFT feature extraction method cosine similarity was used rather than euclidean distance since cosine does not depend on magnitude. Therefore, the distance between a template and a call further away compared to closer together would only be judged on the angular difference and not the difference in magnitude. This ensured that louder and softer calls of the same type could be classified by spectral shape only and not loudness. For the MFCC feature extraction implementation the energy created by the call due to how close or far away it is from the recording device was removed when c_0 was removed as explained in Section 4.2.2. Similarly for LPCC the calculated magnitude is stored within the lower coefficient. This coefficient is always set to 1 during the LPC stage, therefore was not considered as part of the 4 LPC coefficients. This meant that euclidean distance could be used without concern of the effect of magnitude on the distance since only spectral shape is left to be analysed. Euclidean distance is the common choice for DTW analysis, therefore since the main cosine similarity benefit is not necessary for MFCC or LPC, euclidean distance was chosen.[37]

The subseq setting with in the sequence.dtw function was set to True. As explained in Section 2.2, subsequence DTW means that the start and end points of a detected

match do not have to be the first and last points of both signals. When searching 1 hour segments of a 72 hour audio file for whale calls subsequence DTW allows for the detection of multiple calls within a single audio segment by searching the final row of the cost matrix for dips and choosing the lowest point of each dip as an end point of a call. This is further explained in Section4.5. [38]

4.4 Hyperparameter Tuning

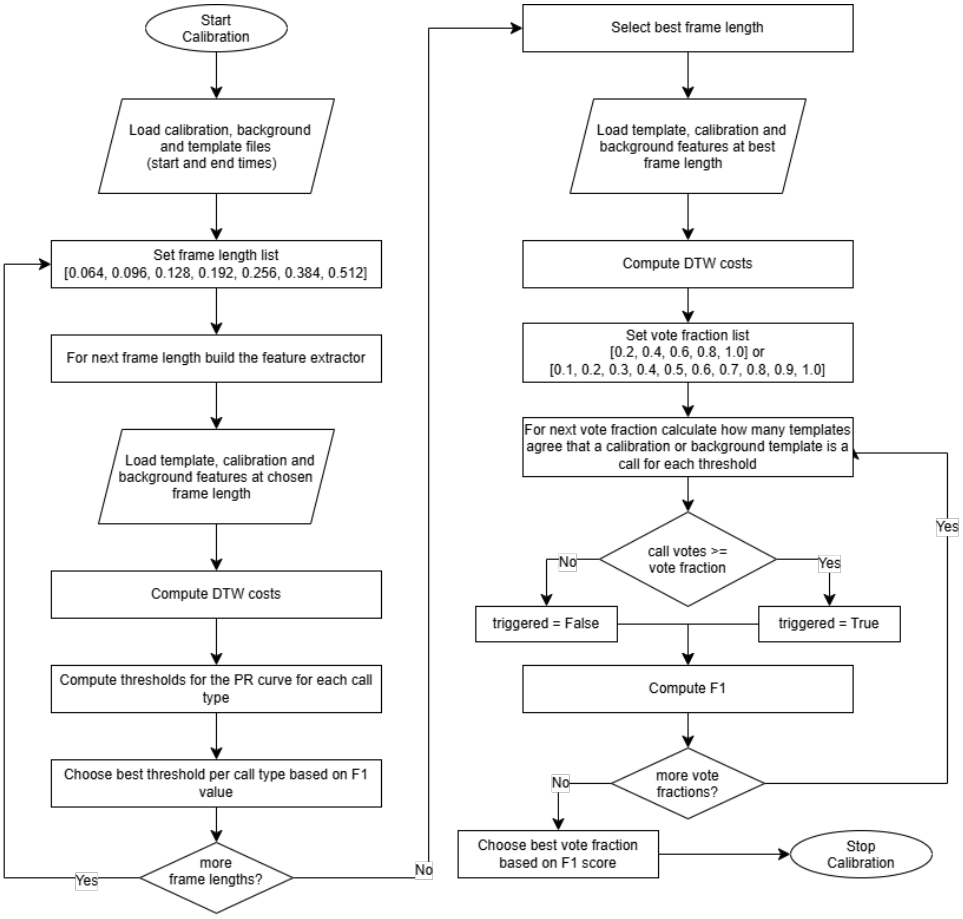


Figure 4.1: This is a software flow diagram describing how the hyperparameters are chosen for the project.

Three hyperparameters were chosen to be tuned, namely feature frame length, call type thresholds and call vote fraction. Figure 4.1 shows the software flow for the hyperparameter selection section of the project.

As shown in Eq 2.9 frequency resolution= $\frac{f_s}{L}$. Considering the fact that Bryde's Whale single-tone calls tend to have a main frequency around 42Hz and multi-tone calls have more than one frequency between 42 and 100 Hz, the upper bound for frequency resolution was chosen as about 10 Hz, this gives a frame length of 100ms. The lower bound in Hz

was chosen as around 2 Hz. While a smaller frequency resolution is better for seperating frequency bins, a resolution of 2 Hz gives a frame length of 500ms. Given a window overlap of 75% this leaves a hop length of 128 ms. Given the average call length within the hyperparameter selection audio file is 246 ms this hop length is still short enough to detect the average call. Since it is computationally best to choose frame lengths which are a power of 2 or multiples of powers of 2 the chosen frame lengths to be tested were (0.096, 0.128, 0.192, 0.384, 0.512).[39]

Since the call type calibration templates and background calibration templates had to be handpicked due to data scarcity and since the subsequence DTW method was used each calibration template was padded on each side. This padding allowed the subsequence DTW to work in conditions similar to those used in the final testing pipeline despite the data restrictions. It was ensured that the padded length was longer than the longest testing template being used with atleast one extra frame length added to each side of the longest template. This extra length was chosen to equal the longest feature frame length tested, 0.512. The call or background calibration template was centered within this length. This was done by moving the start and ends times back and forward respectively to reach this set length.

For each of the tested frame lengths, three thresholds were determined, one for each call type. These thresholds were determined by taking between testing and calibration templates for each call type as well as the DTW costs calculated between testing of each template and the background calibration templates. These costs were stored in two seperate lists, one for call type calibration costs and one for background calibration costs. The lowest and highest costs between the two lists was taken and 200 thresholds were selected between these two numbers in a linearly spaced fashion. The F1 scores were calculated at each threshold for that call type as explained in Section 4.6. The final threshold chosen for the specific call type was the threshold with the highest F1 score.

For each feature frame length there were three thresholds chosen, one for each call type. The lowest F1 score out of these three thresholds F1 scores was saved. These saved F1 scores were then used such that the best feature frame length was chosen as the one where the lowest F1 score between the three thresholds for that frame length was the highest of all the lowest F1 scores between all frame lengths.

While three thresholds were calculated, one for each call type, the project only focused on general detection of calls and not on classification. This is due to the fact that many frequency elements were shared between call types. Even though the three call types are different enough to justify three different thresholds and not one overarching threshold, calls are often misclassified. This choice is also due to the labeling discontinuity since different human labelers had different opinions on what calssified a call as single-tine, multi-tone or burst tonal. This discontinuity meant that the templates selected for each call type were possibly labeled by a different labeler than the one who labeled the evaluation

data. The difference in labeling methodologies could result in calibration thresholds not correctly classifying call types during evaluation.

For the main evaluation pipeline there are 5 or 10 templates against which each audio segment was tested. Since the templates were chosen to represent slight variations in call ypes this meant that while some templates would agree that an audio segment was or was not a call, not all would agree. A vote fraction was used such that only if a certain percentage of the 5 or 10 templates agreed that an audio segment was a call would it be labeled as a call. A list of voting fractions were tested on the best frame length choice to determine the best vote fraction. When 5 templates were used the vote fractions tested were 0.2, 0.4, 0.6, 0.8 and 1.0. When 10 templates were used the vote fractions tested were 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9 and 1.0. The two options is due to the fact that 20% of 5 is 1 and 40% is 2 and so on, thus testing the odd numbered percentages would be unecessary.

For each vote fraction the F1 score for overall call detection was calculated. This means that each threshold was tested against all calibration templates no matter the call type as well as all background calibration templates. This choice is due to the testing pipeline testing all audio segments against all three thresholds. The best vote fraction was chosen as the one with the highest overall F1 score.

4.5 Evaluation Procedure

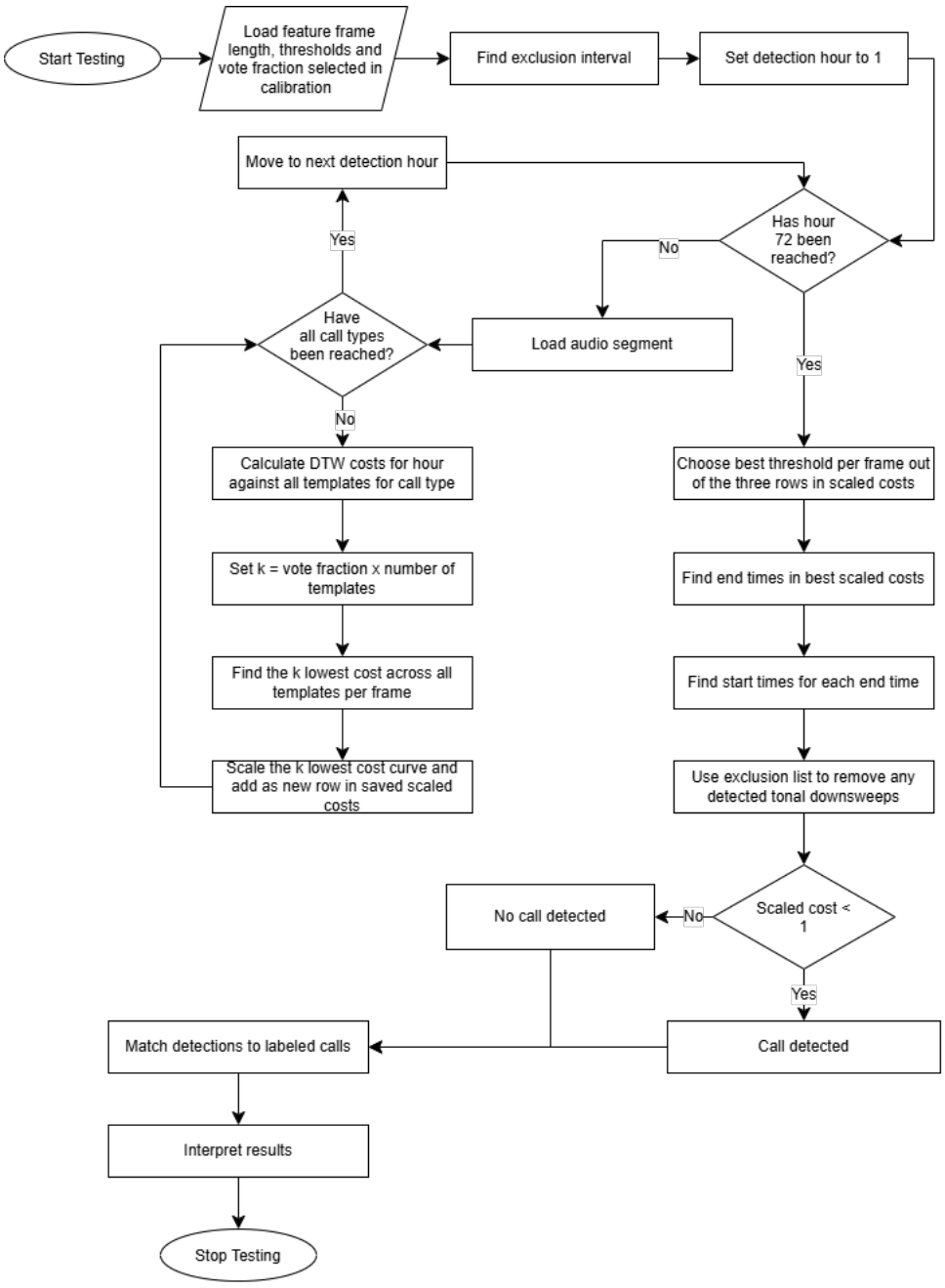


Figure 4.2: This is a software flow diagram describing how the evaluation procedure runs for this project.

4.5.1 Detection Methodology

The variables chosen during the hyperparameter selection stage were used during evaluation. These were the feature frame length, thresholds per call type and the voting fraction. The file used for evaluation was the only file containing tonal downsweeps. Therefore, since no

threshold was calculated for this call type a list of exclusion indices was created for all the labled tonal downsweeps. Due to the use of subsequence DTW the tonal downsweeps were only removed after detecting calls since leaving gaps in time during subsequence DTW could effect the results.

The 72 hour file was processed one hour at a time. This ensured that memory usage was bounded to a smaller amount at one time. The soundfile library was used to load hourly segments of the 72 hour file without running through the entire file at once.[40] The extracted audio was then resampled from 8kHz to 1kHz using the resample_poly function from the SciPy library.[41] Features were extracted as explained in Section 4.2.1.

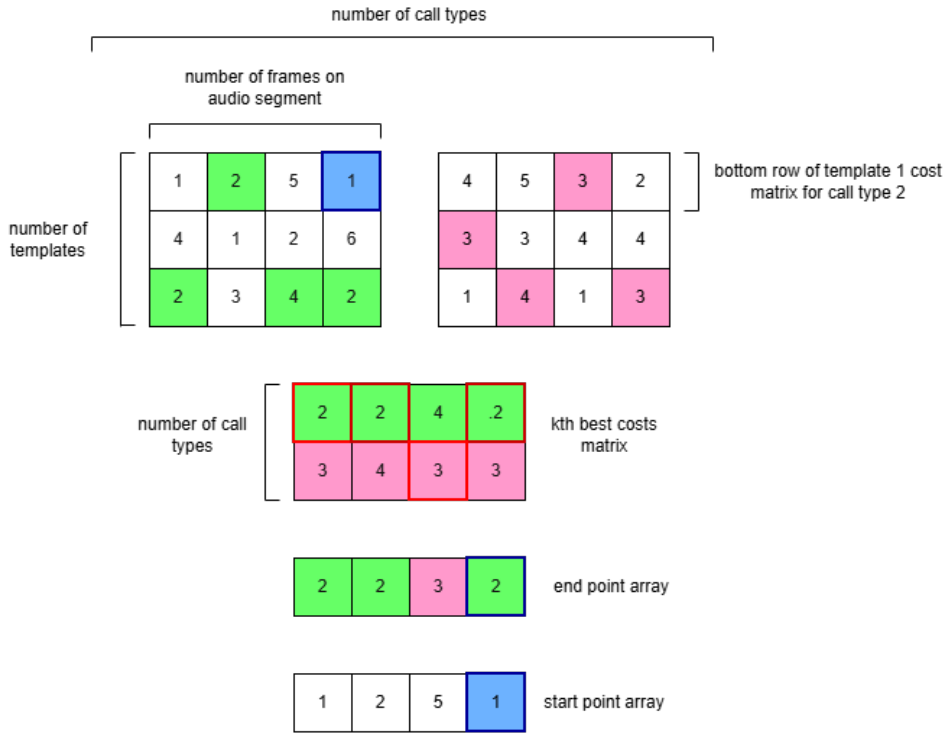


Figure 4.3: This diagram shows how the kth lowest costs are stored for a hypothetical choice of costs per frame for three templates per threshold and two thresholds given k = 2.

The DTW costs for all templates were calculated per call type and the bottom row of each DTW cost matrix for each template was stored in a matrix for every call type. This can be seen in the first image of Figure 4.3. A variable k was calculated to be the vote fraction, decided during hyperparameter selection, multiplied by the number of templates per call type. This variable was used such that the kth lowest cost across all templates, per frame, in the call type matrix was selected and stored in one kth lowest cost array. All the values in the array was divided by the threshold of the given call type. This ensured that a resulting matrix of scaled kth costs per frame could be created such that no matter which threshold the kth cost had been calculated from a cost of 1 would always indicate a

match with the threshold. This can be seen in the second image in Figure 4.3. From the scaled matrix containing the kth best costs per threshold the overall best cost per frame was selected by choosing the lowest stored cost per frame.

This cost was stored and analysed for call detection as shown in the third image in Figure 4.3. A call would only be detected if the percentage of templates for a sepecific call type detecting a call was equal to the vote percentage. Only the k lowest cost had to be considered since any value lower than k would also have a lower cost. Thus if the kth cost detected a call all lower calls would also detect a call and the vote fraction would be met. If, however, the kth cost did not detect a call even if any of the lower cost templates detected a call the vote fraction would not be met, thus a call would not be detected.

The find_peaks function from the SciPy library was used to detect end times within the scaled kth cost 1D-array. Subsequence DTW works in such a way that the end point of a matching sequence is the lowest cost within the last row of the cost matrix. This means that within a 1D-array of a 1 hour segment with many calls there will be various dips in cost which will reach an absolute minimum cost at the end point of a possible call. If the costs considered were to be multiplied by negative one these end points would be found at peaks instead of dips. Thus any place wher $cost >= -1$ would indicate a detected end point of a call.[42]

For any detected end frame, within the scaled kth cost array the start time is calculated. The end time was found using the kth lowest cost. This cost came from one of the ten templates for one of the thresholds. Out of these ten templates there is also one containing the absolute lowest cost for that end frame. The start time is calculated using that template within which the lowest cost is found. A section of the 1 hour audio segment is extracted such that it ends at the end of the chosen end frame and starts ten seconds before the end frame starts. When matching a template to an audio segment using DTW subsequence the detected call can be longer than the original template. Since there is no indication for how long the detected call could be a max of ten seconds was chosen. This is arbitrarily long for a Bryde's Whale call, but as long as the whale call starts somewhere within those ten seconds the correct start time can be found. The reason for the ten seconds is so that the DTW of the entire 1 hour audio segment does not have to be recomputed to find the warp path used to select the start time. This saves computational time.

The sequence.dtw function from librosa is once again used to calculate the DTW cost matrix of this audio segment. This function has the capability to return the best warp path for the given audio segment, however since the end point was selected on the k lowest cost template and the start time is now being chosenon the absolute lowest cost template, this template might have an end point with a higher cost that the selected end point. This end point may just not have been detected due to the vote fraction. Therefore, the step choice matrix needed to find the warp path is returned instead of the best warp path for

that audio segment. The sequence.dtw_backtracking function from the librosa library is then used to find the warping path. In this function it is possible to force the warp path calculation to start from the chosen end point instead of the best for that audio segment. From that chosen end point the function works backwards though the steps matrix until it finds the start point.[43]

Theoretically whether one finds the start point in the kth lowest template or the best should not make too much of a difference, however in the end the kth template was only used to find the end points due to the vote fraction. After finding those end points finding the most accurate start time is important and since the best template has matched the best to that call it m akes sense to find the start time using the best template instead if the kth best template.

4.5.2 Matching Methodology

For each detected call only the labeled calls for the relevant time period were checked to find overlap. The searchsorted function from the numpy library was used to find the labeled calls overlapping the detected calls. Four 1D-arrays were used: start times for detected calls, end times for detected calls, start times for labeled calls, end times for labeled calls. For each detected call searchsorted was used to find at which index, within the array containing labeled call end times, the detected start time would have to be inserted to keep the end times array sorted from smallest to largest. This index was chosen as lo. The searchsorted function was also used to determine at which index within the start times array for the labeled calls the end time from the detected call would have to be inserted to keep the start times array sorted. This was chosen as hi. These lo and hi variables ensured that only labeled calls which started before the detected call ends and ended after the detected call start were considered. If hi was larger than lo then any indices between these two times were labeled calls touching the detected call.The matching logic was set up in such a way that if one or more labeled calls touched the detected call the call was considered to be a true positive call. So as long as hi was larger than lo that detected call was labelled as matched and the labeled call or calls were labeled as matched. If another call were to also touch an already matched labeled call it would still be matched as general detection was the main goal and not specific time matches.[44]

4.6 Evaluation Metrics

While reciever-operating characteristic (ROC) analysis is more common in detection models, due to the class imbalance between whale calls vs background noise, precision and recall (PR) analysis was chosen as the better method for analysis. This is due to the fact that ROC curves tend to give a false impression as to how a model acts in the presence of

class imbalance. PR curves on the other hand take class imbalance into account and thus give a more accurate reflection of how the model acts. Due to this precision, recall and F1 score was chosen as the method of analysis.The one downside which was considered when interpreting results was that due to the lack of data the calibration data used had a different imbalance to the evaluation data. This meant that it was expected that the resulting chosen PR point would be less optimal than the PR curve could have allowed.

Precision is a value showing how many of the detected whale calls actually touched a label. True positives (TP) are defined as a detected call which overlaps at least one labeled call. False positives (FP) are defined as detected calls which do not overlap with any labeled call. The equation to calculate precision is defined as:

$$presision = \frac{TP}{TP + FP} \tag{4.1}$$

Recall is a value showing how many of the labeled calls were detected by the model. In this case a TP value is defined as a labeled call which has been overlapped at least once by a detected call. False negatives (FN) are the labeled calls which are never overlapped by a detected call. The equations for recall is defined as:

$$recall = \frac{TP}{TP + FN} \tag{4.2}$$

It can be seen that for precision and recall TP has a slightly different definition. This is due to the fact that the model at times could incorrectly detect one labeled call as two different calls and sometimes two labeled calls can be detected as one call. Since the goal of this project is detection of calls and not classification the analysis does not consider the detection of one labeled call as two detected calls to be incorrect, thus leading to one of the detected calls being labelled as FP. Instead any detection overlapping any labeled call and vice versa is considered TP.

The F1 score shows the optimal point on a PR curve. This is due to the fact that as precision and recall increases the F1 score will also increase. Therefore, for all choices in hyperparameter tuning the decision is based on the highest F1 score. The F1 score is also used in the evaluation file to determine the overall performance of models relative to each other. The equation for the F1-score is defined as:

$$F1 = \frac{2PR}{P + R} \tag{4.3}$$

[45][46]

For the evaluation part of the project the average precision (AP) was also calculated. AP was calculated using all point interpolation. The equation for this is:

$$\mathrm{AP}_{\mathrm{all}} = \sum_{n} (R_{n+1} - R_n)\, P_{\mathrm{interp}}(R_{n+1}) \tag{4.4}$$

where

$$P_{\text{interp}}(R_{n+1}) = \max_{\tilde{R}\,:\,\tilde{R}\geq R_{n+1}} P(\tilde{R}) \tag{4.5}$$

[47]

The points for the PR curve, used to calculate the AP, were calculated by sweeping the three thresholds equally up and down by a variable alpha. During the main detection pipeline when the end and start times of the detected calls were found on the scaled kth costs, the threshold below which a call was detected was initially set to 1.5. Before matching the detected calls in the initial evaluation the detected calls cost array was again trimmed such that only the calls falling below the actual threshold, 1, were matched against labels. The extra costs, falling under 1.5, were necessary to later ensure there were enough costs available for a threshold sweep for a PR curve. 100 thresholds were tested between 0,01 and 1,5. The precision and recall was calculated at each of the new thresholds.

Chapter 5

Results

After the final detection and seperate analysis of the three different models an overall analysis must be done to determine which of the feature extraction methods was the best as well as whether dynamic time warping (DTW) works overall as a detection algorithm for Bryde's Whales in False Bay.

5.1 Experiment 1: STFT Features

For the short time fourier transform (STFT) model used for call detection two experiments were conducted. The first considered the results given the use of 5 templates used as the analysis database for both hyperparameter selection and evaluation. The second considered the use of 10 templates, the original 5 with 5 new templates added to increase the database diversity.

Table 5.1: Tabulation of thresholds as well as the respective F1-scores used to choose the best frame length. The average F1-score over all three thresholds per STFT experiment was also given for easier analysis.

Number of Temps	Single Tones		Multi Tones		Burst Tonals		Average F1	Frame Length
	F_1	Threshold	F_1	Threshold	F_1	Threshold		
5	0.682	0.234	0.568	0.256	0.796	0.260	0.682	0.512
10	0.690	0.245	0.663	0.236	0.697	0.230	0.683	0.512

Both the 5 and 10 template experiments had an average F1-score of about 0.68 over all three thresholds. Both chose 0.512 seconds as the best frame length. The individual F1-scores for the 5 template experiment were less consistent across the three thresholds than for the 10 template experiment. This can be seen in Table 5.1 as for the 10 template experiment the F1-score consistently stays between 0.66 and 0.70. For the 5 template experiment the change in F1-score between thresholds was much larger staying between

0.56 and 0.80. The absolute highest F1-score for a single threshold was 0.796 for the burst-tonal call type of the 5 template experiment.

Table 5.2: Vote frac results for STFT for 5 and 10 templates

Number of Temps	Precision	Recall	F1	Vote Fraction
5	0.964	0.900	0.931	0.6
10	0.967	0.967	0.967	0.4

The precision, recall and F1 scores for the two STFT experiments during vote fraction selection can be seen in Table 5.2. The vote fraction for the 5 template experiment was 0.6 and for the 10 template experiment it was 0.4. This means a larger percentage of the 5 template experiment votes had to agree on a call than for the 10 template experiment.

In terms of the precision, recall and F1-score the 10 template experiment performed the best for its chosen vote fraction. The final F1-score was 0.967. This shows that for the chosen thresholds at the best possible feature frame length the general detection of a call when compared to all 30 call type templates, 10 for each call type, performed well under the chosen vote fraction. There is an expected drop in precision, recall and F1-score for detection of evaluation data. This drop is due to the vote fraction test being applied using the same calibration templates used to choose the three thresholds and feature frame length. The reuse of calibration templates is due to data scarcity and thus gives an overly optomistic performance result of the model which should only be considered in choosing the vote fraction and not in determining the actual model performance.

Table 5.3: Final test results for STFT for 5 and 10 templates

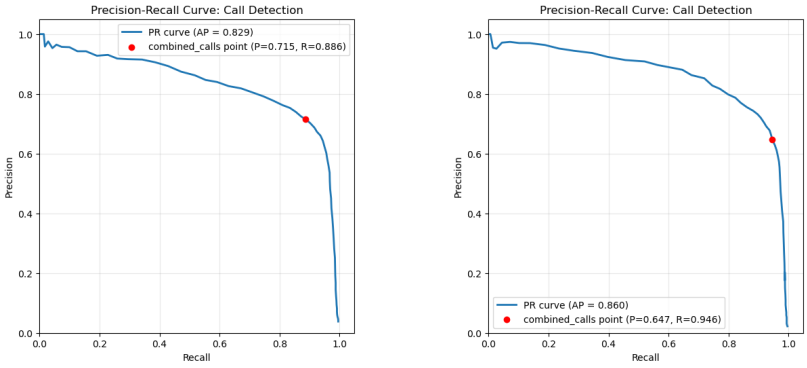
Number of Temps	Precision	Recall	F1	Average Precision
5	0.715	0.886	0.792	0.829
10	0.647	0.946	0.769	0.860

In terms of precision, as seen in Table 5.3, the 5 template experiment performed the best with a difference in precision between the two experiments as 0.068 and the highest being 0.715. However, in terms of recall the 10 template experiment performed the best with a 0.06 difference and the highest being 0.946.

Since the precision difference between the two experiments was the largest the 5 template experiment outperformed the 10 template experiment. This can also be seen in the final F1 score with the highest being 0.792 for the 5 template experiment.

By looking at Figures 5.1a and 5.1b it can be seen that both experiments performed well in terms of the selection of the hyperparameters as both marked precision and recall points

are in the optimal position of their respective curve. Therefore despite the slightly better numerical performance both STFT experiments performed well in terms of hyperparameter selection.



(a) 5 templates **(b)** 10 templates

Figure 5.1: The PR curves for STFT

By again looking at Table 5.1 and Figures 5.1a and 5.1b, It can be seen that in terms of average precision (AP) and therefore the overal precision recall (PR) curve of the model the 10 template experiment outperformed the 5 template experiment with the highest AP equal to 0.860.

5.2 Experiment 2: MFCC Features

For the mel-frequency cepstral coefficient (MFCC) model three main experiments were considered. The first experiment used the MFCC output with no added derivatives for the feature. The second experiment used the original 13 feature coefficients from experiment 1 and added 13 more coefficients in the form of the derivitives of the original 13. The third and final experiment consider the 26 coefficients used in the second experiment and added another 13 in the form of the second derivitives of the original 13 MFCC output coefficients.

For each experiment two sub-experiments were performed. The first was performed using 5 templates and the second using 10 templates.

Table 5.4: Calibration variables chosen via highest F1 for 5 templates and 10 templates

	Number of Temps	Single Tones		Multi Tones		Burst Tonals		Avergage F1	Frame Length
		F_1	Threshold	F_1	Threshold	F_1	Threshold		
MFCC	5	0.718	16.394	0.675	22.688	0.500	19.483	0.631	0.384
	10	0.784	17.507	0.402	19.553	0.472	21.288	0.553	0.256
MFCC	5	0.683	15.610	0.659	23.440	0.500	20.337	0.614	0.384
deriv1	10	0.785	18.815	0.394	22.548	0.425	20.569	0.535	0.256
MFCC	5	0.758	18.199	0.659	24.977	0.518	21.595	0.600	0.384
deriv2	10	0.806	19.642	0.408	22.356	0.430	22.836	0.548	0.256

As seen in Table 5.4 all three experiments for the MFCC model chose the same frame length for the 5 and 10 template sub-experiments. These values were 0.384 and 0.256 seconds for the 5 and 10 template sub-experiments respectively.

For all three experiments the 5 template sub-experiment achieved a higher average F1-score over all three thresholds. In terms of average F1 score over all three thresholds the MFCC experiment without derivatives performed the best. The highest average F1-score was for the 5 template sub-experiment with a score of 0.631. The second highest average F1-score was for the experiment where the first derivitives were added with a score of 0.614.

In terms of the 10 template sub-experiments the experiment only using the first 13 MFCC output coefficients peroformed the best in terms of average F1-score with a score of 0.553. The second best F1-score across the three 10 template sub-experiments was for the third experiment using all 39 coefficients. This score was 0.548.

For all three experiments and their respective sub-experiments the single-tone call type performed the best in terms of F1-score for its chosen threshold. The highest F1-score was 0.806 for the 10 template sub-experiment of the third experiment containing all 39 coefficients

Table 5.5: Final vote results for 5 and 10 templates

	Number of Temps	Precision	Recall	F_1	Vote Fraction
MFCC	5	1.000	0.900	0.947	0.6
	10	0.935	0.967	0.951	0.5
MFCC	5	0.935	0.967	0.951	0.4
deriv1	10	0.964	0.900	0.0.931	0.6
MFCC	5	0.906	0.967	0.935	0.4
deriv2	10	1.000	0.900	0.947	0.6

In terms of the F1-score for the choice of vote fraction, as seen in Table 5.5, the highest score of 0.951 was shared by the 10 template sub-experiment for the MFCC experiment without derivatives and the 5 template sub-experimant for MFCC experiment with the first derivitives added.

The second highest F1-score of 0.947 was shared by the 5 template sub-experiment for the MFCC experiment without derivitives and the 10 template sub-experiment of the MFCC experiment with two derivatives.

Considering that the MFCC experiment with no derivatives held both the first and second highest F1-scores this model was seen to have performed best in terms of overall call detection at the chosen hyperparameter settings for the calibration data. This performance can only be considered in terms of the hyperparameter choices and not in terms of the overall performance of the model due to the use of the same calibration templates for all hyperparameter selection choices.

The vote fractions chosen for the MFCC experiments with the first and second derivitives stayed constant in terms of number of templates. The 5 template sub-experiments chose a vote fraction of 0.4 and the 10 template sub-experiments chose 0.6. The MFCC experiment without derivitives had different vote fractions with the 5 template sub-experiment choosing 0.6 and the 10 template sub-experiment choosing 0.5.

Table 5.6: Final test results for 5 and 10 templates

	Number of Temps	Precision	Recall	F_1	Average Precision
MFCC	5	0.582	0.714	0.641	0.642
	10	0.535	0.817	0.647	0.684
MFCC deriv1	5	0.495	0.799	0.611	0.632
	10	0.468	0.809	0.593	0.602
MFCC deriv2	5	0.354	0.880	0.505	0.614
	10	0.505	0.770	0.610	0.628

For the MFCC experiment with no derivatives the 5 template sub-experiment performed the best for precision. The 10 template sub-experiment performed best in terms of recall and F1 score. The highest values for precision, recall and F1 score respectively were 0.582, 0.817 and 0.647.

For the MFCC experiment with the first derivatives added the 5 template sub-experiment performed best in precision and F1 score while the 10 template sub-experiment performed best for recall. The highest values for precision, recall and F1 score respectively were 0.495, 0.809 and 0.611.

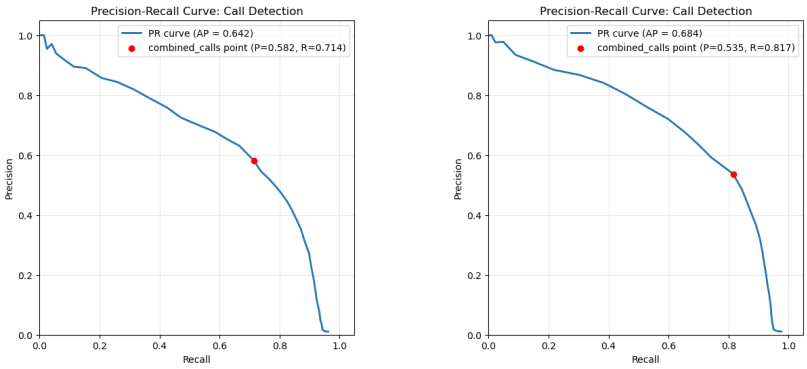
For the MFCC experiment with the second derivatives added the 5 template sub-experiment performed best for recall. The 10 template sub-experiment performed best for

precision and F1 score. The highest values for precision, recall and F1 score respectively were 0.505, 0.880 and 0.628.

Overall in terms of F1-score the 10 template sub-experiment for the MFCC experiment with no derivatives performed the best. By looking at the values in Table 5.6 it can be seen that the highest overall hyperparametr selection performance is characterized by averaging out the precision and recall and not by getting the higest value in either category.

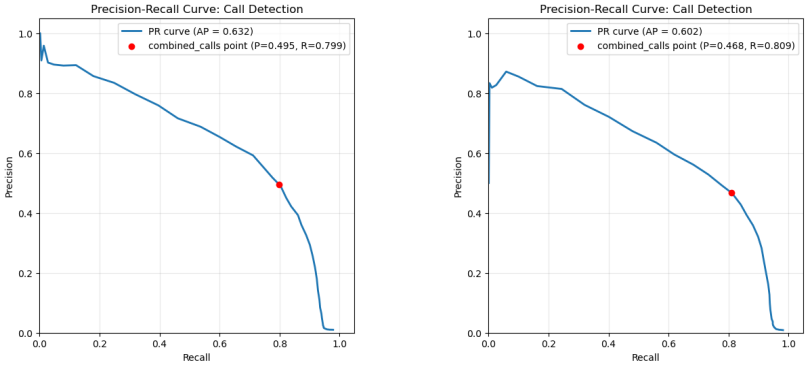
In terms of average precision (AP) and precision and recall (PR) curve the MFCC experiment with no derivatives outperformed all other MFCC experiments with a maximum AP of 0.684 for the 10 template sub-experiment. This can be seen by looking at both Table 5.6 and Figures 5.2, 5.3 and 5.4.

The 10 template sub-experiment for the MFCC experiment with no derivitives is the overall best performer for the MFCC models.



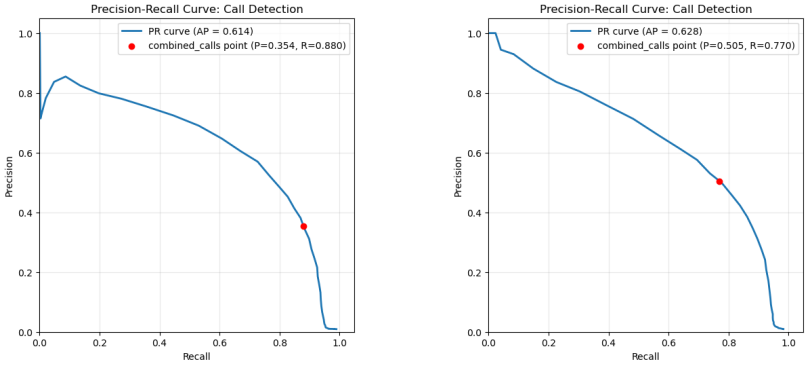
(a) 5 templates **(b)** 10 templates

Figure 5.2: The PR curves for MFCC



(a) 5 templates **(b)** 10 templates

Figure 5.3: The PR curves for MFCC with the first derivative added



(a) 5 templates **(b)** 10 templates

Figure 5.4: The PR curves for MFCC with the first and second derivatives added

5.3 Experiment 3: LPCC Features

For the linear predictive coding coefficients (LPCC) model two experiments were conducted. The first considered the use of 5 templates and the second considered the use of 10 templates for the call type database for hyperparameter selection and evaluation.

Table 5.7: Calibration variables chosen via highest F1 for LPC for 5 templates and 10 templates

Number of Temps	Single Tones		Multi Tones		Burst Tonals		Averag F1	Frame Length
	F_1	Threshold	F_1	Threshold	F_1	Threshold		
5	0.863	0.344	0.744	0.526	0.632	0.449	0.746	0.256
10	0.882	0.336	0.605	0.364	0.686	0.427	0.724	0.256

Both the 5 and 10 template experiments for the LPC model chose 0.256 seconds as the frame length. This can be seen in Table 5.7.

The 5 template model had the highest average F1-score of 0.746. This score was 0.022 higher than the to 10 template experiment F1-score. There was no consistency in terms of F1-scores across the three call type thresholds for either the 5 or 10 template experiment.

Table 5.8: Vote frac results for LPC for 5 and 10 templates

Number of Temps	Precision	Recall	F1	Vote Fraction
5	0.968	1.000	0.984	0.4
10	0.968	1.000	0.984	0.5

While the 5 and 10 template experiments for LPC each chose a different vote fraction, in terms of precision, recall and F1 score they performed equally well. The chose vote fraction for each was 0.4 and 0.5 for the 5 and 10 template experiments respectively. The F1-score for both experiments was 0.984. This can be seen in Table 5.8.

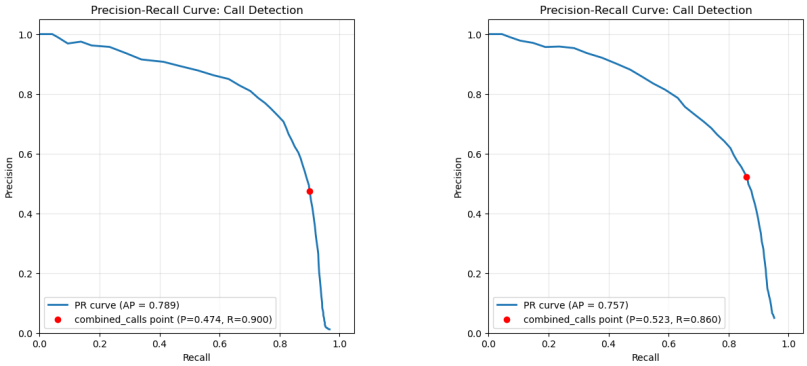
This F1-score carries no weight in terms of the actual performance of the model when tested against evaluation data.

Table 5.9: Final test results for LPC for 5 and 10 templates

Number of Temps	Precision	Recall	F1	Average Precision
5	0.474	0.900	0.621	0.789
10	0.523	0.860	0.650	0.757

As seen in Table 5.9, the 5 template experiment performed better in terms of recall. The 10 template experiment performed best in precision and F1 score. The highest precision, recall and F1-score were 0.523, 0.900 and 0.621 respectively.

This means that the 10 template experiment performed better in terms of chosen thresholds. This can also be seen in Figures 5.5a and 5.5b where the marked precision and recall point is better matched to the precision recall (PR) curve for the 10 template experiment.



(a) 5 templates **(b)** 10 templates

Figure 5.5: The PR curves for LPC with the first and second derivatives added

By considering Table 5.9 and Figure 5.5a and 5.5b, it can be seen that in terms of average precision (AP) and PR curve shape the 5 template experiment performed better with an AP value of 0.789.

Even though the 10 template experiment gave a better PR point due to chosen thresholds the 5 template experiment for the LPC model performed best overall in terms of Bryde's Whale call detections due to the PR curve shape and AP value.

5.4 General Observations

The lack of data and human error meant that a maximum of 10 templates per call type could be used for call type calibration templates. It also caused a bias in the choices of templates to train on as selecting suitably representative templates per call type became difficult. The selection of background calibration templates meant that the class imbalance between calls and background noise differed between hyperparameter selection and evaluation. The selection of templates and not a continuos audio file for hyperparameter selection also meant that subsequence dynamic time warping (DTW) did not necessarily act within the same conditions as evaluation thus creating another difference between the two datasets.

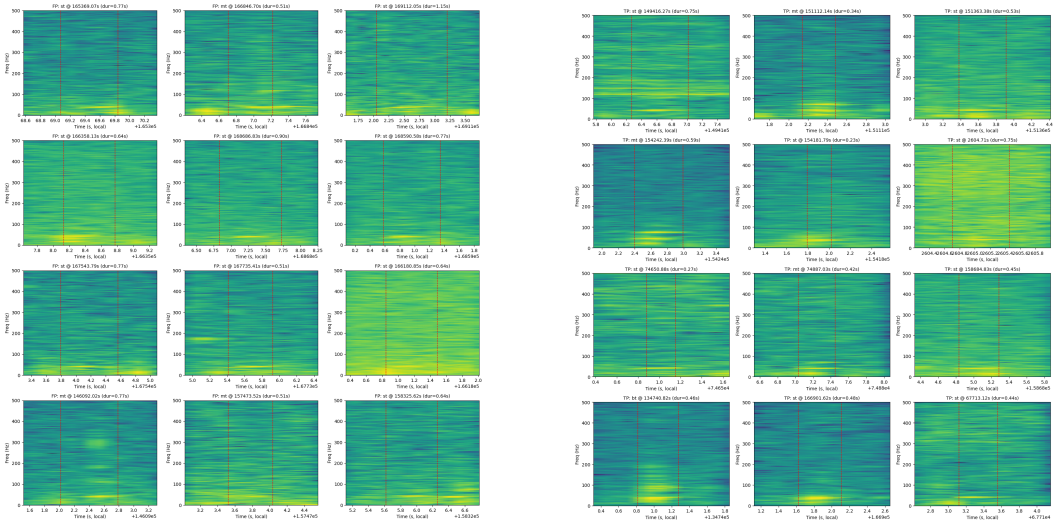
Considering the low availability of well labeled data to use for the hyperparameter selection stage of this project as well as human error, which was introduced in manual selection of calibration templates, the varying performance of the calibrated thresholds is justified. This varying performance can be seen in Tables 5.1, 5.4 and 5.1.

Table 5.10: Overall highest results for F1 and AP per model

Feature extraction method	F1		AP	
	Highest value	Number of Templates	Highest value	Number of Templates
STFT	0.792	5	0.860	10
MFCC	0.647	10	0.684	10
LPC	0.650	10	0.789	5

By looking at Table 5.10 it can be seen that the short time fourier transform (STFT) model outperforms the other models in terms of both F1-score and average precision (AP). This meant that both in terms of the models ability to select hyperparameters and in terms of the efficiency of the detection pipeline the STFT model outperformed the other two models. The linear predictive coding coefficient (LPCC) model comes second, but in terms of precision it is not much better than the mel-frequency cepstral coefficient (MFCC) model.

When looking at printed spectrograms of false positives (FP) detected by the STFT model it was found that for both experiments for the best performing model many of these FP calls were actually unlabeled calls. This means that the actual precision of the model may be higher than the reported amounts. This may be due to the human error introduced in call labeling, since the known labeled calls were found by previous projects and was therefore not necessarily a complete list. Examples of FP calls compared to similar true positive (TP) calls can be seen in Figure 5.6

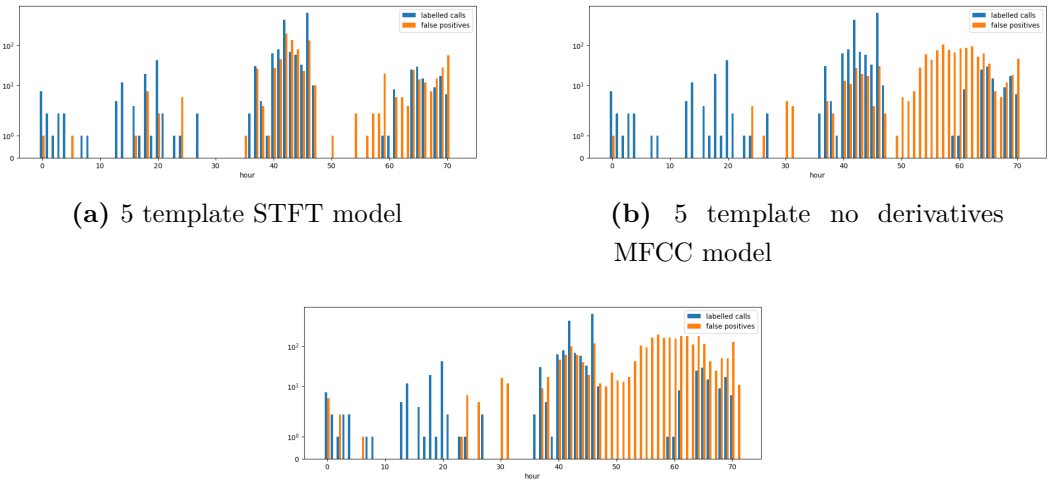


(a) False Postive samples

(b) True Positive samples

Figure 5.6: Examples of random samples of FP and TP for which some FP could be unlabeled calls for the 10 template experiment for the STFT model.

Another interesting observation across all three detection models was that most of the FP calls were found within the second half of the 72 hour audio file used for the evaluation stage of this project. This implies a possible link between model performance and the signal-to-noise ratio (SNR) of the section of audio considered. This observation can be seen in Figure 5.7



(a) 5 template STFT model

(b) 5 template no derivatives MFCC model

(c) 5 template LPCC model

Figure 5.7: This shows for each model where FP calls were detected in relation to labeled calls over the entire 72 hour audio file.

Considering the good performance of the model as well as the unlabeled TP calls, the STFT feature extraction is the best method to use for call detection with the DTW

algorith and the algorithm is a valid choice as detection method for Bryde's Whales within the False Bay region.

Chapter 6

Conclusion

References

[1] K. van Waerebeek, A. N. Baker, F. Félix, J. Gedamke, M. Iñiguez, G. P. Sanino, E. Secchi, D. Sutaria, A. van Helden, and Y. Wang, "Vessel collisions with small cetaceans worldwide and with large whales in the southern hemisphere; building a standardized database," 2006, https://www.vliz.be/imisdocs/publications/243193.pdf.

[2] A. Faustmann, F. W. Shabangu, and J. D. J. J. Versfeld, "Potential vocalizations of bryde's whales in false bay, south africa," 2024, https://repository.up.ac.za/server/api/core/bitstreams/57ba6a31-7aca-4acb-b9b6-60884695c964/content.

[3] S. Betz, K. Bohnsack, A. R. Callahan, L. E. Campbell, S. E. Green, and K. M. Labrum, "Reducing the risk of vessel strikes to endangered whales in the santa barbara channel," 2011, https://nmssanctuaries.blob.core.windows.net/sanctuaries-prod/media/archive/protect/shipstrike/pdfs/bren.pdf.

[4] V. B. Deecke, J. K. B. Ford, and P. Spong, "Call diversity in the north pacific killer whale populations: implications for dialect evolution and population history," *Animal Behaviour*, 2000, https://www.sciencedirect.com/science/article/abs/pii/S0003347211005537.

[5] M. A. McDonald, S. L. Mesnick, and J. A. Hildebrand, "The geographic variation of blue whale song," 2003, https://www.cetus.ucsd.edu/docs/reports/McDonaldIWC-2003.pdf.

[6] J. C. Brown, A. Hodgins-Davis, and P. J. O. Miller, "Classification of vocalizations of killer whales using dynamic time warping," 2006, https://pubs.aip.org/asa/jasa/article/119/3/EL34/850621.

[7] J. Herreilers, C. Jacobs, and T. Niesler, "Low-resource keyword spotting using contrastively trained transformer acoustic word embeddings," 2025, https://arxiv.org/pdf/2506.17690.

[8] E. M. Oleson, J. Barlow, J. Gordon, S. Rankin, and J. A. Hildebrand, "Low frequency calls of bryde's whales," 2003, https://digitalcommons.unl.edu/cgi/viewcontent.cgi?referer=&httpsredir=1&article=1297&context=usdeptcommercepub.

[9] M. Müller and F. Zalkow, "Fundamentals of music processing: Dynamic time warping (dtw)," 2015, https://www.audiolabs-erlangen.de/resources/MIR/FMP/C3/C3S2_DTWbasic.html.

[10] H. Kamper, "Dynamic time warping (dtw)," 2021, https://www.youtube.com/playlist?list=PLmZlBIcArwhMJoGk5zpiRlkaHUqy5dLzL.

[11] R. Schwaber-Cohen, "Vector similarity explained," 2023, https://www.pinecone.io/learn/vector-similarity/.

[12] E. Alizadeh, "An illustrative explanation to dynamic time warping," 2020, https://towardsdatascience.com/an-illustrative-introduction-to-dynamic-time-warping-36aa98513b98/.

[13] M. Müller and F. Zalkow, "Fundamentals of music processing: Subsequence dtw," 2015, https://www.audiolabs-erlangen.de/resources/MIR/FMP/C7/C7S2_SubsequenceDTW.html.

[14] D. J. J. V. Atinde M. Usman, Olayinka O. Ocundile, "Review of automatic detection and classification techniques for cetacean vocalization," 2020, https://scholar.sun.ac.za/server/api/core/bitstreams/6e4a861d-74eb-4d2d-b81b-353743cc06b8/content.

[15] H. Kamper, "Short-time fourier transforms (stft)," 2026, https://www.youtube.com/watch?v=tQeDi2qEW-s&list=PLmZlBIcArwhMrUZ_njMmlO_T8gAWHihNS&index=26.

[16] ——, "Introduction to speech features," 2021, https://www.youtube.com/playlist?list=PLmZlBIcArwhN8nFJ8VL1jLM2Qe7YCcmAb.

[17] T. Singh, "Mfcc's made easy," 2019, https://medium.com/@tanveer9812/mfccs-made-easy-7ef383006040.

[18] DataMListic, "Spectral features - deltas and delta-deltas explained," 2023, https://www.youtube.com/watch?v=zxEnuPolylY.

[19] ——, "Mel frequency cepstral coefficients (mfcc) explained," 2023, https://www.youtube.com/watch?v=SJo7vPgRlBQ.

[20] S. A. Majeed, H. Husain, S. A. Samad, and T. F. Idbeaa, "Mel frequency cepstral coefficients (mfcc) feature extraction enhancement in the application of speech recognition: A comparison study," 2015, https://www.jatit.org/volumes/Vol79No1/5Vol79No1.pdf.

[21] L. Rabiner and B.-H. J. Juang, *Fundamentals of Speech Recognition*. Prentice Hall, 1993.

[22] J. C. Brown and P. J. O. Miller, "Automatic classification of killer whale vocalizations using dynamic time warping," 2007, https://pubs.aip.org/asa/jasa/article-abstract/122/2/1201/699104/Automatic-classification-of-killer-whale?redirectedFrom=fulltext&casa_token=OzamuY0etVkAAAAA:VNGCuA1SmTnQEYgVRN2KuKcJvbDF79bLT2VkmggZSIUc2qnhYfNt_Yo7BleyPL4ffIKE3CTopQ.

[23] O. O. Ogundile and D. J. J. Versfeld, "Analysis of template-based detection algorithms for inshore bryde's whale short pulse calls," 2020, https://ieeexplore.ieee.org/document/8957435.

[24] X. Mouy, M. Bahoura, and Y. Simard, "Automatic recognition of fin and blue whale calls for real-time monitoring in the st. lawrence," 2009, https://pubs.aip.org/asa/jasa/article-abstract/126/6/2918/627068/Automatic-recognition-of-fin-and-blue-whale-calls?redirectedFrom=fulltext&casa_token=_uPiaa9QXCsAAAAA:Ge3i-ujMkyY8PsWGIQ4iGgW0FRxsPD0VtXtwuCMONoJvGUJFQAuBTQu8Iq7TWFSif-LMp-Eos

[25] O. O. Ogundile, A. Usman, O. Babalola, and D. Versfeld, "A hidden markov model with selective time domain feature extraction to detect inshore bryde's whale short pulse calls," 2020, https://www.sciencedirect.com/science/article/pii/S1574954120300376?casa_token=6Nq5H72ZrPQAAAAA:X0axwemDJS8TgltJ62x5518DbviKiMtXRjb7HX_TJkOnpH1IpbmheAeQcZnjfbrx6TNF-oP_Z7o.

[26] Y. Permanasari, E. H. Harahap, and E. P. Ali, "Speech recognition using dynamic time warping (dtw)," 2019, https://iopscience.iop.org/article/10.1088/1742-6596/1366/1/012091/meta.

[27] A. Faustmann, "Potential vocalizations of bryde's whales in false bay, south africa," 2024, https://scholardata.sun.ac.za/articles/dataset/An_Analysis_of_Potential_Low_Frequency_Vocalizations_of_Southern_African_Bryde_s_Whales_/25108016?file=44351747.

[28] B. Meddins, "The nyquist frequency," 2000, https://www.sciencedirect.com/topics/computer-science/nyquist-frequency.

[29] A. M. Learning, "Windowing functions explained," 2026, https://apxml.com/courses/introduction-to-speech-recognition/chapter-2-processing-audio-signals/windowing-functions-explained.

[30] T. S. community, "dct," 2008, https://docs.scipy.org/doc/scipy/reference/generated/scipy.fft.dct.html.

[31] M. A. Roch, A. Širović, and S. Baumann-Pickering, "Detection, classification, and localization of cetaceans by groups at the scripps institution of oceanography and san diego state university," 2013, https://www.cetus.ucsd.edu/docs/publications/RochDCL2013.pdf.

[32] librosa development team, "librosa.feature.mfcc," 2026, https://librosa.org/doc/main/api/generated/librosa.feature.mfcc.html.

[33] ——, "librosa.feature.delta," 2023, https://librosa.org/doc/dev/api/generated/librosa.feature.delta.html.

[34] B. McFee, C. Raffel, D. Liang, D. P. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in python," 2015, https://proceedings.scipy.org/articles/Majora-7b98e3ed-003.pdf.

[35] librosa development team, "librosa.lpc," 2025, https://librosa.org/doc/0.11.0/generated/librosa.lpc.html.

[36] ——, "librosa.util.frame," 2025, https://librosa.org/doc/0.11.0/generated/librosa.util.frame.html.

[37] T. S. community, "scipy.spatian.distnace," 2008, https://docs.scipy.org/doc/scipy/reference/generated/scipy.spatial.distance.cdist.html.

[38] librosa development team, "librosa.sequence.dtw," 2026, https://librosa.org/doc/0.11.0/generated/librosa.sequence.dtw.html.

[39] P. Hees, "The fast fourier transform, how and why it works," 2026, https://research.ibm.com/blog/the-fast-fourier-transform-how-and-why-it-works.

[40] B. Bechtold, "soundfile 0.14.0," 2026, https://pypi.org/project/soundfile/.

[41] T. S. community, "resample_poly," 2008, http://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.resample_poly.html.

[42] ——, "find_peaks," 2008, https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.find_peaks.html.

[43] librosa development team, "Source code for librosa.sequence," 2026, https://librosa.org/doc/latest/_modules/librosa/sequence.html.

[44] N. Developers, "numpy.searchsorted," 2026, https://numpy.org/doc/stable/reference/generated/numpy.searchsorted.html.

[45] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the roc plot when evaluating binary classifiers on imbalanced datasets," 2015, https://www.researchgate.net/publication/273155496_The_Precision-Recall_Plot_Is_More_Informative_than_the_ROC_Plot_When_Evaluating_Binary_Classifiers_on_Imbalanced_Datasets.

[46] T. Fawcett, "An introduction to roc analysis," 2005, https://www.sciencedirect.com/science/article/pii/S016786550500303X.

[47] R. Padilla, S. L. Netto, and E. A. B. da Silva, "A survey on performance metrics forobject-detection algorithms," 2020, https://www.researchgate.net/publication/343194514_A_Survey_on_Performance_Metrics_for_Object-Detection_Algorithms.

Appendix A

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